

World's CHEAPEST And SMALLEST VENTILATOR Made In India



Nidhi Arora is
executive editor
at EFY

Ventilators have become synonymous for emergency care. Unfortunately, due to their high cost, not every hospital and patient in the world can afford them. The good news is that the world's smallest and cheapest ventilator has been made in India, revolutionising critical care. This ventilator can be purchased by all levels of hospitals from primary care hospitals to tertiary care hospitals. The scope of its commercialisation is quite huge.

The journey

The new ventilator is the result of a joint effort between Professor Diwakar Vaish, head, robotics and research, A-SET Training & Research Institutes, and Dr Deepak Agrawal, professor-neurosurgery at AIIMS, New Delhi.

"We met during a programme in AIIMS and there we discussed various challenges that our healthcare system is facing currently. Out of all the problems, we centered on ven-

tilators, due to lack of which precious lives are being lost. From there on we decided to work together to make a cheaper and more efficient ventilator. Our main aim was to create something that is cheap, easy to run and portable," shared Prof. Vaish.

The technology was developed by A-SET with medical inputs from AIIMS.

Challenges

Conventional ventilators are very expensive (costing ₹ 1-1.5 million each), extremely big (1-1.5m tall and around 61cm wide and deep) and complicated, requiring medical staff to run them. In addition, these need constant oxygen supply to run, which costs around ₹ 2000-3500 per day.

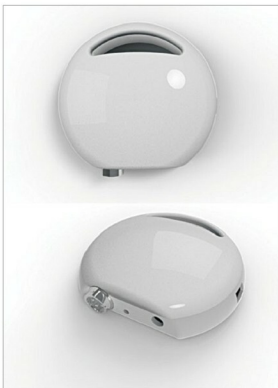
A-SET Robotics' ventilator overcomes these problems as:

1. It is around a hundred times cheaper at ₹ 15,000.
2. It has the size of a compact disk, so it can fit into the user's pocket.
3. It doesn't require oxygen supply and can circulate room air to the patients.
4. No professional expertise is required to control it. It can be operated by just a smartphone via Bluetooth interface, which makes it extremely easy to use. Also, in case of any problem, one may reach out to the provided helpline numbers, and it can be controlled remotely by the central team.

How it works

A machine learning algorithm understands the patient's inhale-exhale pattern and controls air pressure and flow rate to the patient as per the best settings:

1. A pressure sensor monitors the relative pressure of the patient's air passage over 1000 times a minute
2. The pressure reading is then sent to a computing unit, which calculates if it's inhale phase or exhale phase
3. In inhale phase, the air is induced at a controlled pressure and sent at



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